Investigating Latent Reasoning in Multimodal Models

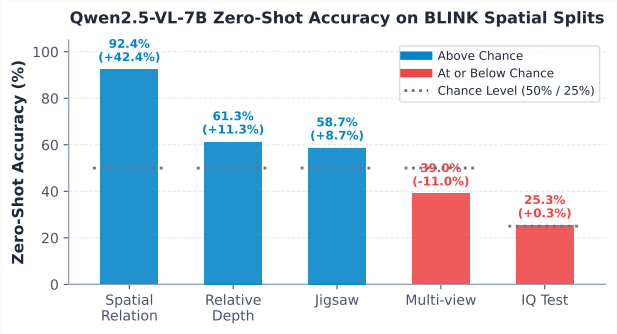
Does "Thinking" in Continuous Hidden States Truly Help Vision-Language Models See?

Vanshika

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Vision-Language Research Group · `MTok Project`

1. Why Am I Doing This? (The Problem)



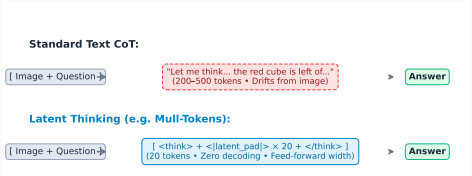
The Perception vs. Reasoning Gap

- **Object Recognition is Solved**: Spatial Relation = **92.4%** (+42% over chance).
- **Spatial Manipulation Collapses**: Multi-View = **39.0%** *(below 50% chance)*
  IQ-Test = **25.3%** *(at 25% chance)*

Why Text CoT Fails Here

Language is a lossy channel for space: Writing thoughts in words (Text CoT) causes **visual drift** and actually **hurts performance** (−3 pp vs. direct answering).

2. What Am I Doing? (The Core Question)



The Proposed Fix: Latent Tokens

Instead of decoding words, prefill 20 identical dummy tokens:

`<think> + <|latent_pad|> × 20 + </think>\n → answer`

Claim: Model computes spatial relations internally in hidden layers without verbal drift (+5.4% gain).

My Research Question: Do Latents Compute?

- **Hypothesis 1 (Scratchpad)**: Latents compute image features. (Perturbing them should hurt accuracy).
- **Hypothesis 2 (Format Key)**: Latents do no computation; the prefix simply acts as a **syntactic format stop-gate**.

Every paper reports accuracy gains. Almost nobody audits what the latents actually do!

3. How Am I Doing This? (Replication First)

1. Fixed 5 Upstream Codebase Bugs

- **Chat Template Corruption**: Upstream script stripped all turn boundaries, silently mangling prompts.
- **Hardcoded Paths**: Task YAMLs pointed to unreleased cluster directories $\rightarrow$ rewired to HuggingFace.
- **Model Registration & Schemas**: Fixed missing decorators and dataset schema mismatches.

2. Rigorous Data Integrity Controls

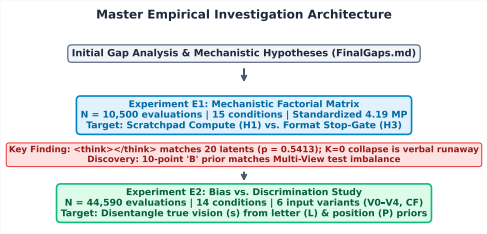
- Automated assertions enforce exact sample counts ($N = 700$ total) to catch smoke-test contamination.
- Full transparent forensic log kept in `ERRATA.md`.

3. Full Replication on A100 GPU ($N \approx 7,500$)

Benchmark Split	Paper Claim	Our Replication
BLINK Spatial Rel. (143)	81.8%	**85.3%** (+3.5%)
BLINK Rel. Depth (124)	83.1%	**85.5%** (+2.4%)
BLINK Jigsaw (150)	74.0%	**76.7%** (+2.7%)
BLINK Multi-View (133)	63.2%	**63.9%** (Matched)
BLINK IQ-Test (150)	32.0%	**30.7%** (Matched)
BLINK Avg (700)	**66.8%**	**68.4%** (+1.3%)
VSI-Bench Video (5,130)	33.0%	**33.7%** (+0.7%)

Published results replicated and exceeded; testbed verified.

3. How Am I Doing This? (Mechanistic Design)



Experiment E1: Factorial Matrix ($N = 10,500$)

15 controlled conditions across all 5 tasks:

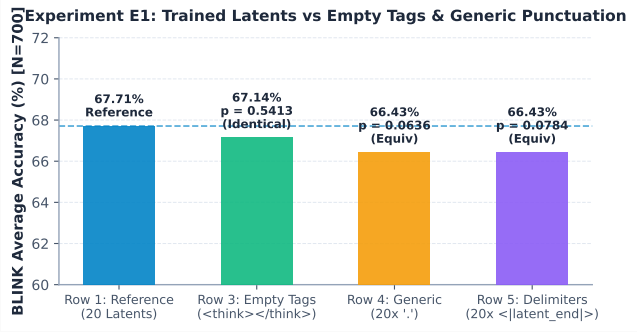
- Negative controls: Empty tags (`<think></think>`), punctuation (20x `'.'`), delimiters.
- Budget scaling: $K \in \{0, 1, 5, 10, 20, 40\}$.
- Standardized 4.19 MP resolution with paired McNemar tests.

Experiment E2: Bias vs. Vision ($N = 44,590$)

14 conditions $\times$ **6 controlled input variants**:

- Real (V0), Blind Grey Canvas (V1), Option Swap (V2), Relabel (V3), Swap+Relabel (V4), Content-Free Dummy (CF).
- Mathematical debiasing: Extracts pure visual score s free of letter (L) and position (P) biases.

4. What Results Did I Get? (Finding 1: Zero Latents Match)



The Decisive Proof ($N = 700$)

Row 3 (Empty Tags: `<think></think>`):

- Injects **ZERO** latent tokens.
- Scores **67.14%** vs. Reference **67.71%** ($\Delta = -0.57\%$).
- Paired McNemar: $b = 14, c = 10$, $p = 0.5413$ (**Statistically Identical!**)

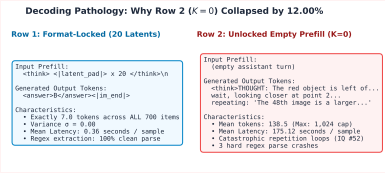
Punctuation Controls Match

- 20 Periods (`"."`): **66.43%** ($p = 0.0636$)
- Delimiter tokens: **66.43%** ($p = 0.0784$)

Takeaway: Trained weights carry **no specialized visual reasoning payload**.

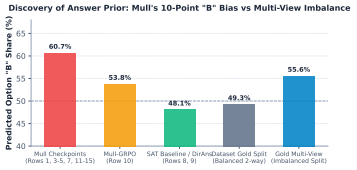
4. What Results Did I Get? (Finding 2: The Two Confounders)

1. Why $K = 0$ Dropped 12% (Decoding Runaway)



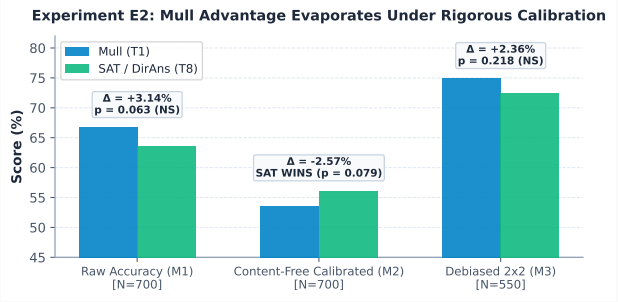
- When `<think>` is unclosed, model falls into unconstrained verbal text-CoT.
- Latency blows up **480**× (0.36s → **175.1s/sample**).
- IQ-52 loops 48 cycles hitting 1,024 cap; 3 items fail regex parse.
- **Proof (T2F)**: Forcing direct `<answer>` with 0 latents scores **66.86%** (matches full latents!).

2. Severe 10-Point "B" Answer Prior



- Mull predicts option **"B" 60.7%** of the time on 2-way tasks (DirAns baseline is calibrated at 48.1%).
- BLINK Multi-View split is skewed: **55.6% gold "B"**.
- Guessing "B" gives a free artificial score boost!

5. Result Analysis: Does Mull Beat Direct Answering?



Primary Hypothesis Q1 ($N = 700$)

- **Raw Accuracy ($M1$)**: Mull 66.71% vs. SAT 63.57% $\Delta = +3.14\%$ ($p = 0.063$, non-significant).
- **Content-Free Calibrated ($M2$)**: Mull 53.57% vs. **SAT 56.14%** **Direct Baseline WINS by +2.57%** ($p = 0.079$).
- **Debiased 2×2 Discrimination ($M3$)**: Mull 74.91% vs. SAT 72.55% $\Delta = +2.36\%$ ($p = 0.218$, non-significant).

Core Discovery: **35.0% of Mull's reported margin is pure answer-prior alignment** with imbalanced test splits!

5. Result Analysis: The Scientific Verdict

What Mull-Tokens Actually Are

Mull-Tokens are a **syntactic format stop-gate**, NOT a continuous scratchpad or world model.

- Prefilling `<think>...</think>` locks the assistant turn and terminates generation.
- It protects the model from discursive verbal CoT, preventing catastrophic visual drift.
- Continuous logit margins: $W = 46497.0$, $p = 0.9833$ (mathematically identical to empty tags).

Independent Convergence with 2026 Audits

Four recent papers concurrently found latents collapse:

- *Viveiros et al. (NeurIPS 2026)*: Dummy tokens match latents.
- *Guo et al. (Aug 2026)*: Boundary markers give 78–100% of gain.
- *Li et al. (ICML 2026) & Zhang et al. (EMNLP 2026)*.

Our Unique Contribution:

- Audited Mull as the **limiting case** (discrete identical tokens, zero slot content).
- Discovered the 35% answer-prior confound.
- Included full video spatial evaluation (VSI-Bench).

5. What Further From Here? (Next Steps)

Proposed Direction: Adaptive Latent Allocation

Key Insight: Content-free tokens provide format control and parallel FLOPs, but a fixed 20-token prefix cannot allocate compute where it is actually needed.

- **1. Task Structure Taxonomy (Weeks 1–3)**: Map exactly which spatial & video tasks benefit from format locking vs. decoder FLOPs using layer attention knockout.
- **2. Adaptive Budget Controller (Weeks 4–7)**: Train a lightweight LoRA to learn dynamic per-instance stopping (`<|end_of_thought|>`) instead of static $K = 20$.
- **3. Conference Submission (Weeks 8–10)**: Target top-tier venue (NeurIPS / CVPR).

Feasibility & Compute Guarantee

- **Current Infrastructure**: Phases 1 & 2 run entirely on our existing **1× A100-40GB GPU**.
- **Targeted Compute Request**: 1 week of 80GB VRAM access during Phase 3 to scale GRPO group size ($G = 8/16$) and eliminate dead updates.

Summary Status for Professor

Replication Verified ✓
Mechanistic Audit Definitive ✓
Data Hygiene Rock-Solid ✓
Ready to Build Adaptive Controller →